

PRODUCT DATA SHEET

BRSMMH — POSITIVE ELECTRIC ADHESIVE DRY POWDER

Positively charged mixed-layered metal hydroxide (MMH) crystal colloid for drilling fluid stabilization

Description

BRSMMH is a positive electric adhesive dry powder — a positively charged mixed-layered metal hydroxide (MMH) crystal colloid — developed by Brighter Star Drilling Fluids together with partner research institutes through extensive laboratory testing and field practice, formulated specifically for HDD applications. The positively charged MMH crystals adsorb onto negatively charged clay and cuttings surfaces, forming a strong composite network structure that delivers unique shear-thinning rheology, high yield stress and exceptional suspended cuttings-carrying capacity. BRSMMH functions both as a drilling fluid stabilizer and a wellbore stabilizer, and is especially suited to complex formations, hard rock and horizontal wells.

BRSMMH is best at increasing the dynamic shear force of drilling fluid, improving the dynamic plastic ratio, and greatly enhancing the carrying capacity. It is particularly suitable for rock formations that are prone to producing large rock debris, which can maximize the carrying of drilling debris, reduce secondary cutting, lower drilling tool wear, and greatly improve drilling efficiency.

Executive standard: Q/PSL038-2003.

Main Functions

- Increases structural viscosity and yield stress of the drilling fluid
- Stabilizes the wellbore and inhibits shale hydration and swelling
- Provides exceptionally strong suspended rock cuttings carrying capacity
- Delivers shear-thinning flow behavior for smooth, efficient circulation
- Helps increase drilling rate and reduce overall drilling costs



Product Features

- ✓ Purely inorganic, environmentally friendly
- ✓ Zeta potential ≥ 35 mV, strong clay adhesion
- ✓ Yield stress improvement rate $\geq 300\%$
- ✓ Proven in rock formations and horizontal wells

Packaging

Packaging Form	Net Weight	Options
Triple-layer paper sack with PE inner liner	25 kg per bag	Customized packaging on request

Storage

Store in a cool, dry, clean and well-ventilated place at 5–40 °C (do not exceed 40 °C) and relative humidity $\leq 80\%$. Keep containers tightly closed to prevent moisture ingress; protect from direct sunlight, heat

sources and moisture. Do NOT store together with toxic, harmful or polluting substances, or with strong oxidizing agents. Shelf life: 12 months from the date of manufacture under proper sealed-storage conditions.

Application & Dosage

BRSM MH® is suitable for freshwater-based drilling systems in HDD, horizontal wells and complex formations. Reference dosage ranges (by weight of water):

- As viscosifier / stabilizer: 0.5% – 0.8%
- As shale inhibitor: 0.5% – 1.0%

The reference dosages above are based on general practice for positively charged MMH products. The actual dosage shall be determined by on-site pilot tests according to well depth, formation characteristics and the drilling fluid system.

1. Prepare fresh water and check its quality; an untreated bentonite base slurry is preferred. Avoid adding anionic additives (e.g. polyphosphates, lignosulfonates) before BRSM MH®, as they weaken the positive-charge network.
2. Pre-mix BRSM MH® with a small amount of clean water into a uniform slurry.
3. Add the slurry slowly through the hopper into the circulating tank under continuous agitation.
4. Circulate fully for 1–2 cycles, then test the rheological parameters (yield stress, gel strength, zeta potential) and fine-tune the dosage accordingly.

Product Specification

Executed per enterprise standard Q/PSL038-2003.

Item	Specification
Appearance	White or light brown free-flowing powder
Loss on Drying	≤ 15.0 %
Screen Residue (40-mesh)	≤ 7.0 %
Potential (Zeta)	≥ 35.0 mV
Acid Solubility	≥ 85.0 %
Yield Stress Improvement Rate	≥ 300.0 %



Physical Properties

Property	Typical Value
Product Type	Positively charged mixed-layered metal hydroxide crystal colloid
Physical Form	Free-flowing dry powder
Colour	White to light brown
Charge Character	Positively charged (cationic)
Transport Classification	Not regulated as dangerous goods

All data, recommendations and information contained in this Brighter Star Drilling Fluids publication are provided for reference only. They are derived from laboratory testing and field experience obtained under specific conditions, and actual product performance may vary with

formation characteristics, water quality, equipment and operating practices. Users are advised to verify the suitability of this product through on-site trials and to adjust the dosage according to actual working conditions. No warranty as to accuracy, reliability or completeness, either expressed or implied, is given other than those required by law.